

Vikas Narang

Research Associate

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Summary

- Research Associate and PhD candidate focused on agriculture robotics, with emphasis on navigation and controls in unstructured environments.

Education

PhD	Arizona State University , Systems Engineering	Expected Dec 2027
	• Relevant courses: Robotics Systems I, Robotics Systems II, Mobile Robotics, Optimization and Controls, Modern Controls, Reinforcement Learning, Digital Signal Processing, Digital Controls, Engineering Mathematics, Probability and Stochastic Processes, and Vehicle Dynamics and Controls.	
M.S.	Michigan Technological University , Mechanical Engineering	MI, USA 2012
B.S.	M.D. University , Mechanical Engineering	Rohtak, India 2006

Experience

	Arizona State University , Research Associate, Dynamics, Systems and Controls Lab	2024-2025
	• Conducting research on teleoperation and motion control for autonomous vehicles using GPS, camera, onboard sensing, and drive-by-wire platforms. • Designing and evaluating control strategies with predictive perception for lateral and longitudinal vehicle operation in remote driving environments. • Developing control-oriented models to analyze the impact of communication latency in 4G/5G-based vehicle teleoperation systems. • Exploring latency-compensation approaches to improve safety, stability, and responsiveness in connected autonomous vehicle teleoperation.	
	Cummins Inc. , Sr. Technical Specialist, Simulation, Controls and Analytics	2021-2024
	• Led development of ML-based solutions for hybrid and conventional powertrain systems, including diagnostic algorithms, resulting in \$14M estimated combined operational and fuel savings. • Coordinated PoC projects across analytics and controls teams to reduce downtime by 15%. • Performed techno-economic analyses of ICE/HEV platforms using fleet telematics data. • Served as SME on data-driven modeling, data strategy, and system behavior alignment.	

Cummins Inc , Technical Specialist, Advanced Dynamic Systems & Controls • Led controls development and validation for spark-ignited engines, including natural gas and propane platforms, supporting combustion stability, fuel control, air-path management, and system robustness. • Lead calibration development using GT-Power simulations, test-cell data, and vehicle testing to validate air-path and fuel-system control algorithms, improving torque response, combustion stability, transient performance, and drivability. • Collaborated with TPMs and cross-functional teams to deliver validated control features from concept through simulation, test-cell evaluation, and vehicle deployment. • Applied VOC and FMEA methods to guide control algorithm development and ensure system-level quality, safety, and robustness.	2016-2020
Cummins Inc , System Performance & Integration Engineer, Advanced Systems Integration • Planned and executed performance testing for diesel and spark-ignited engines in test-cell and vehicle environments. • Performed combustion, performance, and emissions analysis to support engine calibration, fuel economy improvement, aftertreatment integration, and product readiness for on-highway natural gas powertrain programs. • Used GT-Power simulations and system-level models to evaluate engine performance, air-path behavior, fuel-system response, thermal behavior, and integration tradeoffs across operating conditions. • Lead test-cell calibration and validation activities, including steady-state and transient testing for combustion stability, emissions compliance, torque response, and system robustness. • Led vehicle road-trip and field validation activities for Cummins diesel and natural gas engine platforms, supporting data collection, issue resolution, calibration feedback, and real-world performance evaluation. • Defined and managed vehicle and system-level requirements by working closely with controls, calibration, applications, test, and product engineering teams.	2012-2016

Patents

- US Patent 11,002,202 - Deep Reinforcement Learning for air handling control.
- US Patent 10,746,123 - Deep Reinforcement Learning for fuel system referencing.
- US20240326838A1 / EP4408718A1 / WO2023056007A1 - Optimized diagnostics using vehicle data.

Publications

- SAE (2019) - LPG Direct Injection for Medium Duty Trucks.
- ASME (2024) - Deep Q-learning for Plug-in Hybrid Energy Management.

Technical Skills

Tools: MATLAB, Simulink, Python, CarSim, GT-Power, Azure Databricks, Git, ROS 2

Certifications

- ADAS
- Intro to Self-Driving Cars

Organizations

- SAE: WCX Co-Chair; Reviewer - Vehicle Dynamics, Safety, Emissions.
- ASME: Reviewer - Dynamic Systems, Autonomous Vehicles, Energy Systems.
- IEEE: Reviewer - Dynamic Systems and Controls.